

FINAL PROJECT REPORT
FarmPest Guide: A Dangerous Farm Insect Classification System



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ABSTRACT

Dangerous agricultural insects pose a serious threat to crop productivity and quality, requiring fast and accurate identification methods to support effective pest management. This study aims to compare and evaluate three pretrained deep learning architectures, namely EfficientNet-B3, ResNet-50, and InceptionV3, in classifying dangerous farm insects using a computer vision approach. The dataset used consisted of 1591 insect images with 15 classes, which were processed through preprocessing and model training stages using transfer learning techniques. Performance evaluation was carried out based on classification reports, confusion matrices, and Grad-CAM visualizations to analyze accuracy, error patterns, and model interpretability. The results showed that EfficientNet-B3, which was fine-tuned on the top 30 layers, provided the best performance with an accuracy of 0.79 and more stable generalization capabilities compared to other models. The best model was then implemented in the FarmPest Guide application as a classification system for dangerous agricultural insects to support more efficient and sustainable agricultural practices.

I. INTRODUCTION

1. Background

Harmful insects pose a significant threat to agricultural productivity by damaging crops and reducing yield quality. Accurate identification of these insects is crucial for effective pest management and the selection of appropriate control methods. Traditionally, insect identification relies on expert knowledge and manual observation of morphological features, which can be time-consuming and impractical for large-scale agricultural environments. With the rapid development of computer vision and machine learning technologies, automated image-based insect identification has emerged as a promising solution to support modern and sustainable farming practices.

2. Problem Statement

The traditional process of insect identification still relies on manual observation by humans, which is time-consuming and difficult to apply on a large agricultural scale. Therefore, the main problem in this study is how to develop a computer vision-based system that is capable of classifying dangerous farm insects directly, accurately, and efficiently to support more effective pest management.

3. Objectives

This study aims to compare and evaluate three pretrained deep learning model architectures, namely EfficientNet-B3, ResNet-50, and InceptionV3, in classifying dangerous farm insects by looking at the results of the classification report analysis, confusion matrix, and Grad-CAM interpretation to assess the performance. The best model results will be used for the FarmPest Guide application.

4. Significance

This research has practical significance by providing a deep learning-based classification system for dangerous farm insects that can help identify pests more quickly and accurately to support more effective agricultural management. Academically, this research provides comparative insights into the performance of different pretrained CNN architectures, which can serve as a reference for future research in the development of computer vision-based insect classification systems.

II. RELATED WORKS

No.	Paper Title	Research Objective	Methodology	Result	Limitation
1.	Exploring multiple optimization algorithms in transfer learning with EfficientNet models for agricultural insect classification	To evaluate different optimization algorithms for EfficientNet-based insect classification using transfer learning.	EfficientNetB0–B7 models trained on 1,591 insect images with multiple optimizers and analyzed using Grad-CAM.	EfficientNetB3 with AdamW achieved the best performance with 95.5% accuracy.	CNNs trained from scratch performed poorly, and some misclassifications remained.
2.	Comparison of CNN Models With Transfer Learning in the Classification of Insect Pests	To identify the most effective CNN transfer learning model for insect pest classification.	Transfer learning using Xception, MobileNetV2/V3, DenseNet-201, and InceptionV3 on the IP-102 dataset.	DenseNet-201 with fine-tuning achieved the highest accuracy of 70%, outperforming other tested models.	Limited exploration of hyperparameter tuning and deployment efficiency.
3.	Detecting and Classifying Household Insects in Iraq by using Transfer Learning Models	To develop a detection and classification system for household insects using deep learning and machine learning methods.	YOLOv8 for detection, ResNet50 and VGG19 for feature extraction, and SVM for classification. (6,548 images).	ResNet50 achieved 95.11% accuracy, outperforming VGG19	Manual labeling and evaluation are limited to a small number of insect species.
4.	Self-supervised learning improves classification of agriculturally important insect pests in plants	To evaluate whether self-supervised learning (SSL) can enhance insect pest classification, especially with limited labeled data.	SSL methods (NNCLR, BYOL, Barlow Twins) on ResNet models; tested on datasets like IP102. Image segmentation is used to enhance	SSL yields higher accuracy and better generalization than supervised learning, especially with scarce	ResNet dependence: SSL requires more computation and longer training.

			feature extraction.	labels.	
5.	Recognition of Common Insect in Field Based on Deep Learning	To develop an automatic insect recognition system that reduces reliance on expert knowledge.	Transfer learning with pretrained CNNs (VGG16/19, Inception). Dataset: 9 insect classes. Augmentation includes basic methods and DCGAN.	High performance (best 97.39% with VGG19). Transfer learning improves convergence and reduces overfitting.	Small dataset; controlled conditions; long training time for full models.

III. METHODOLOGY

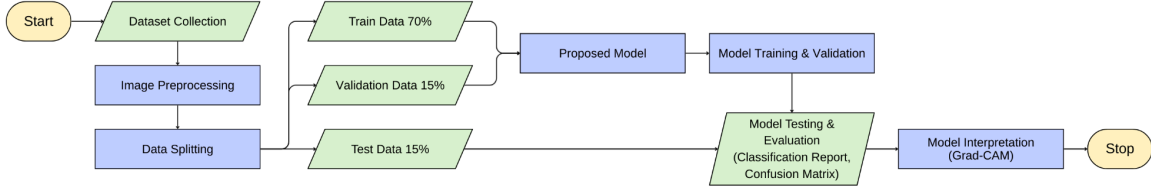


Figure 1. Methodology flowchart.

1. Dataset

The dataset used in this study is the *Dangerous Farm Insects* dataset obtained from Kaggle and uploaded by Tarun R Jain in 2023 under a CC0 Public Domain license. The dataset consists of 1591 labeled images representing 15 classes of agricultural insect pests that commonly threaten crop production. The images exhibit variations in background, lighting, and insect appearance, making the classification task more challenging. The purpose of this dataset is to facilitate research in pest identification, crop protection, and the development of automated computer vision systems for early detection of harmful insects in agriculture.

2. Preprocessing

The preprocessing steps performed include removing broken images, handling duplicate images both within the same class and across different classes, adjusting brightness levels, image sharpening, and resizing all images to 300×300 pixels. After that, each image is also processed using the `preprocess_input()` function according to the model used, namely EfficientNet-B3, ResNet-50, and InceptionV3.

3. Pretrained Model Selection

In this project, we compared 3 pretrained models that performed well in previous research on dangerous farm insect classification, namely:

a) EfficientNet-B3

EfficientNet-B3 is a CNN model that optimizes accuracy and computational efficiency by using a compound scaling approach on depth, width, and resolution. After experimentation, EfficientNet-B3 achieved a train accuracy of 0.8446 and a train loss of 0.5089, with a val accuracy of 0.7569 and a val loss of 0.7817 at epoch 40.

b) ResNet-50

ResNet-50 is a 50-layer deep CNN model that uses residual connections to overcome the vanishing gradient problem in training very deep networks. After experimentation, ResNet-50 achieved a train accuracy of 0.9466 and a train loss of 0.2273, with a val accuracy of 0.7294 and a val loss of 0.8894 at epoch 31.

c) InceptionV3

InceptionV3 is a CNN architecture that uses Inception modules (a combination of parallel 1×1 , 3×3 , and 5×5 convolutions) to capture features at various scales efficiently, then reduces the number of parameters through techniques such as factorized convolutions and dimensionality reduction. After experimentation, InceptionV3 achieved a train accuracy of 0.9276 and a train loss of 0.2983, with a val accuracy of 0.6789 and a val loss of 0.9813 at epoch 37.

Based on these results, we selected the EfficientNet-B3 pretrained model for our application because it is more stable and achieves the best validation performance, with a validation accuracy of 0.7569 and a validation loss of 0.7817.

4. Model Architecture

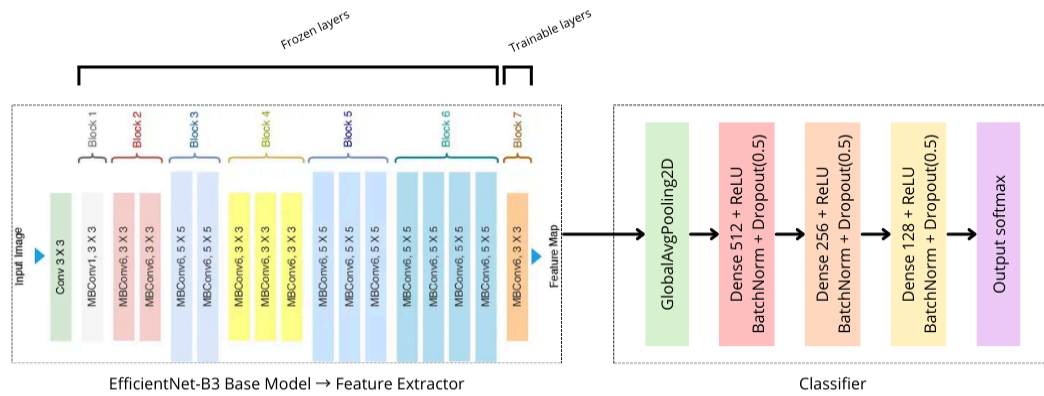


Figure 2. Proposed EfficientNet-B3 model architecture with custom classifier.

a) Pretrained Model: EfficientNet-B3 (Feature Extractor):

Each block of EfficientNet-B3 uses MBConv, which consists of three stages: Expansion, which multiplies channels to enrich features; Depthwise Conv, which processes each channel independently; and Projection, which

reduces the channels again for efficiency. The following are the functions of each block that work on EfficientNet-B3:

- i) Block 1: Early Feature Extraction (Low-Level Features)
Serves to capture basic features such as edges, lines, and low textures. MBConv1 without expansion, thus focusing on efficiency and initial stability.
- ii) Block 2-4: Mid-Level Feature Extraction (Intermediate Semantic Processing)
All blocks use MBConv6 with 6x expansion to create more channels and enrich features.
- iii) Block 5-6: High-Level Feature Extraction (Class-Specific Patterns)
Its receptive field and depthwise conv kernel are larger, enabling it to understand more global visual contexts and capture highly complex and abstract features.
- iv) Block 7: Final Feature Aggregation (Deep High-Level Features)
Expansion, depthwise, and projection from MBConv6 produce high-quality abstract features, which are then converted into a final feature map that will be sent to the classifier.

Blocks 1–6 will be frozen to maintain weight so that they continue to function as fixed feature extractors without change during training. The top 30 layers (primarily within Block 7) are unfrozen and treated as trainable layers, allowing the model to undergo fine-tuning to adapt its high-level features to the dataset and improve task-specific classification performance.

b) Custom Classifier

- i) GlobalAveragePooling2D
Convert the 3D feature map to 1D, take the average of each channel, and retain the most important information.
- ii) Dense 512 + ReLU + BatchNorm + Dropout 0.5
Taking abstract features and then converting them into a more compact representation. BatchNorm for stabilizing the activation distribution. Dropout to prevent overfitting.
- iii) Dense 256 + ReLU + BatchNorm + Dropout 0.5

Re-process features from the previous layer and clarify the separation between classes.

- iv) Dense 128 + ReLU + BatchNorm + Dropout 0.5

Refining features to make them easy to map to output.

- v) Output Dense (Softmax)

Converting the representation into probabilities for each insect class and softmax activation ensures that the total probability is 1.

5. Training Setup

The dataset will be divided into 70% training data, 15% validation data, and 15% test data. All images will be resized to 300×300 pixels with a batch size of 32. Before training, each batch is preprocessed using the appropriate `preprocess_input()` function based on the model being trained so that the image format and normalization comply with ImageNet standards. The training dataset will be shuffled by 1000 samples so that the model does not learn data sequence patterns, and the entire dataset will use `prefetch (AUTOTUNE)` to speed up the pipeline and reduce bottlenecks during training.

The model will use the AdamW optimizer with a learning rate of $1e-4$ and weight decay of $5e-6$, as well as the `sparse_categorical_crossentropy` loss function. Then, class weight is used to handle class imbalance so that the model is not biased. In addition, the `EarlyStopping` and `ReduceLROnPlateau` callbacks are also used to prevent model stagnation.

6. Evaluation Metrics

The model was evaluated using a classification report (accuracy, precision, recall, and F1-score) to see the performance of each class, a confusion matrix to observe the number of correct and incorrect predictions per class, and Grad-CAM to see the parts of the image that the model focused on when making predictions.

IV. IMPLEMENTATION & RESULTS

1. System Details

The experiment was conducted using Visual Studio Code software. The hardware used was a system with a 12th Gen Intel Core i7-12650H (2.30 GHz) processor, 16 GB RAM (4800 MT/s), and a GPU with 4 GB VRAM capacity. The model was trained for a total of 40 epochs with a batch size of 32.

2. Experiments

	Train Acc	Val Acc	Test Acc	Train Loss	Val Loss	Test Loss
EfficientNet-B3	0.8446	0.7569	0.7941	0.5089	0.7817	0.6576
ResNet-50	0.9466	0.7294	0.7605	0.2273	0.8894	0.8277
InceptionV3	0.9276	0.6789	0.7059	0.2983	0.9813	0.9680

3. Visualizations

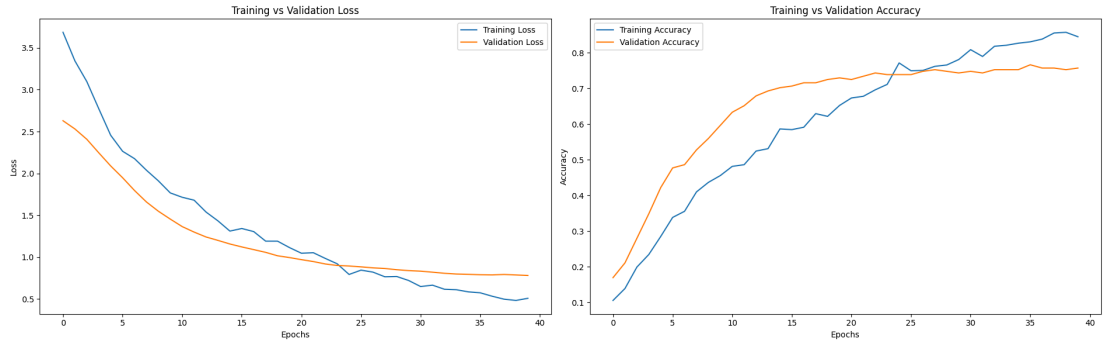


Figure 3. Training vs validation loss and training vs validation accuracy of the EfficientNet-B3 model.

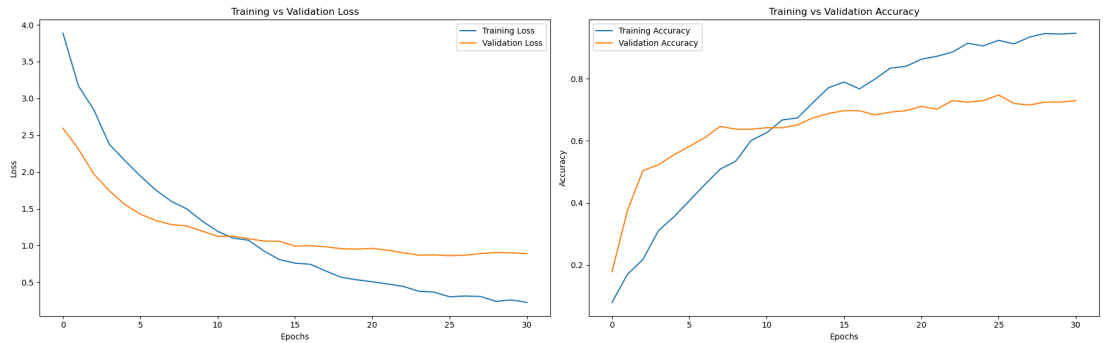


Figure 4. Training vs validation loss and training vs validation accuracy of the ResNet-50 model.

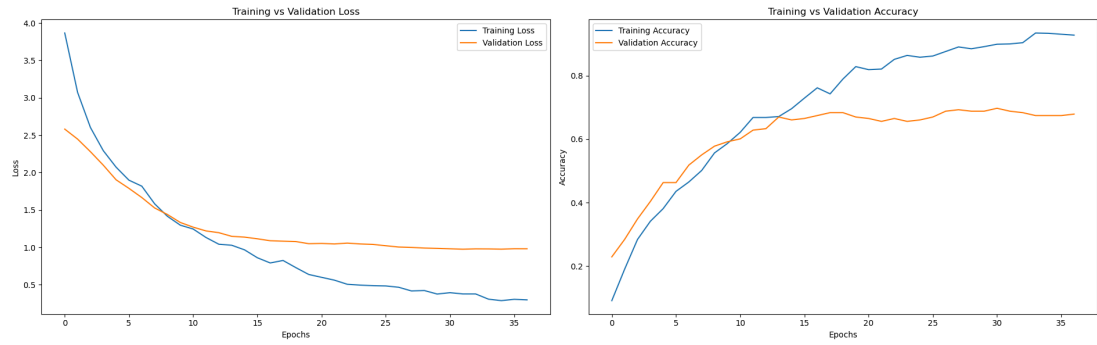


Figure 5. Training vs validation loss and training vs validation accuracy of the InceptionV3 model.

4. Result Obtained

Classification Report:					Classification Report:					Classification Report:				
	precision	recall	f1-score	support		precision	recall	f1-score	support		precision	recall	f1-score	support
Africanized Honey Bees (Killer Bees)	1.00	0.93	0.97	15	Africanized Honey Bees (Killer Bees)	1.00	0.87	0.93	15	Africanized Honey Bees (Killer Bees)	0.93	0.93	0.93	15
Aphids	0.77	0.77	0.77	13	Aphids	0.83	0.77	0.80	13	Aphids	0.67	0.62	0.64	13
Armyworms	0.38	0.45	0.42	11	Armyworms	0.62	0.73	0.67	11	Armyworms	0.30	0.27	0.29	11
Brown Marmorated Stink Bugs	1.00	0.83	0.91	18	Brown Marmorated Stink Bugs	0.88	0.78	0.82	18	Brown Marmorated Stink Bugs	0.93	0.72	0.81	18
Cabbage Loopers	1.00	0.76	0.87	17	Cabbage Loopers	0.86	0.53	0.67	17	Cabbage Loopers	0.89	0.47	0.62	17
Citrus Canker	0.84	1.00	0.91	16	Citrus Canker	1.00	0.88	0.93	16	Citrus Canker	0.84	1.00	0.91	16
Colorado Potato Beetles	0.94	0.94	0.94	17	Colorado Potato Beetles	1.00	0.94	0.97	17	Colorado Potato Beetles	0.83	0.88	0.86	17
Corn Borers	0.60	0.67	0.63	18	Corn Borers	0.53	0.56	0.54	18	Corn Borers	0.65	0.61	0.63	18
Corn Earworms	0.73	0.65	0.69	17	Corn Earworms	0.52	0.65	0.58	17	Corn Earworms	0.53	0.47	0.50	17
Fall Armyworms	0.20	0.36	0.30	14	Fall Armyworms	0.42	0.57	0.48	14	Fall Armyworms	0.29	0.36	0.32	14
Fruit Flies	0.93	0.93	0.93	15	Fruit Flies	0.93	0.93	0.93	15	Fruit Flies	0.93	0.93	0.93	15
Spider Mites	0.94	0.89	0.91	18	Spider Mites	0.88	0.83	0.86	18	Spider Mites	0.76	0.72	0.74	18
Thrips	0.78	0.82	0.80	17	Thrips	0.75	0.71	0.73	17	Thrips	0.71	0.59	0.65	17
Tomato Hornworms	1.00	1.00	1.00	17	Tomato Hornworms	0.68	0.88	0.77	17	Tomato Hornworms	0.70	0.94	0.80	17
Western Corn Rootworms	0.85	0.73	0.79	15	Western Corn Rootworms	0.80	0.80	0.80	15	Western Corn Rootworms	0.61	0.93	0.74	15
accuracy			0.79	238	accuracy			0.76	238	accuracy			0.71	238
macro avg	0.80	0.78	0.79	238	macro avg	0.78	0.76	0.77	238	macro avg	0.71	0.70	0.69	238
weighted avg	0.82	0.79	0.80	238	weighted avg	0.79	0.76	0.77	238	weighted avg	0.72	0.71	0.70	238

Figure 6. Classification reports for: (a) EfficientNet-B3, (b) ResNet-50, and (c) InceptionV3.

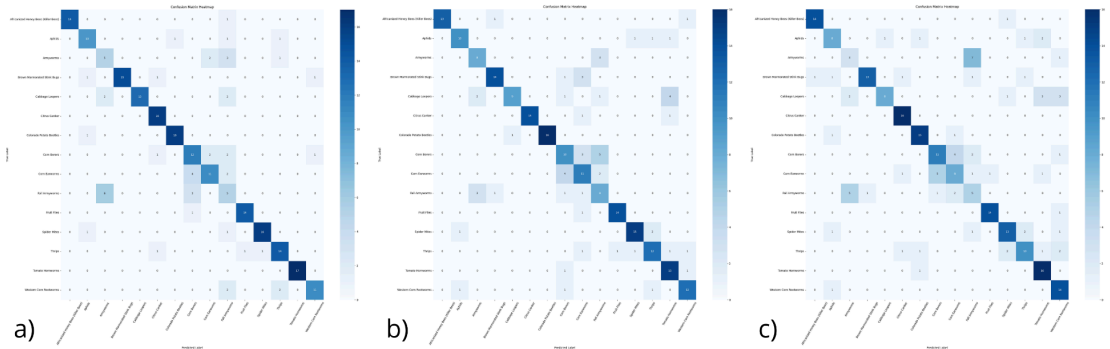


Figure 7. Confusion matrix for: (a) EfficientNet-B3, (b) ResNet-50, and (c) InceptionV3.

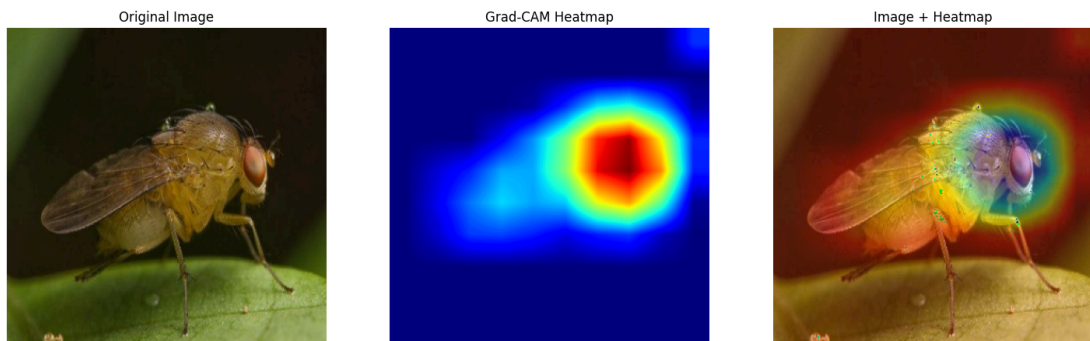


Figure 8. Example of Grad-CAM visualization for the Fruit Flies class using EfficientNet-B3.

V. DISCUSSION & LIMITATIONS

1. Analysis of Performance

Based on Figures 3, 4, and 5, the results on the train, validation, and test sets show that the EfficientNet-B3 model with a custom classifier performs the best due to its good generalization ability, stability, and lack of significant overfitting. In comparison, the ResNet-50 and InceptionV3 models show more overfitting, albeit slightly, due to the larger gap between training and validation and testing.

From the classification report results in Figure 6, the EfficientNet-B3 model has the best performance with an accuracy of 0.79, followed by ResNet-50 with an accuracy of 0.76, and finally InceptionV3 with an accuracy of 0.71. From the confusion matrix in Figure 7, it can be seen that the three models experience the same error pattern, especially in Corn Borers, Corn Earworms, and Fall Armyworms, which are interchanged because they have similar body shapes and image backgrounds. In addition, Armyworms are also often confused with Fall Armyworms because they are still in the same genus, so their patterns and body textures are similar. These four classes produced low F1-scores across all models.

Overall, the difference in performance between the three models was influenced by the suitability of the architecture to the size and characteristics of the dataset. EfficientNet-B3 excelled because its compound scaling design was able to capture visual details more efficiently, as seen from its stable F1-score in almost all classes. Although ResNet-50 has skip connections that aid training stability, its capacity is less than optimal for complex insect shape variations, resulting in performance below EfficientNet-B3. Meanwhile, InceptionV3 showed the lowest performance because its more complex architecture was not fully absorbed by the small dataset size.

2. Challenges

- a) The small size of the dataset, consisting of only 1,591 images, limits the visual variation of each class, making it difficult for the model to learn rich features. This has an impact on classes with similar morphology or those from closely related genera, increasing the risk of prediction errors.

- b) Computational limitations due to the entire training process being carried out using personal devices without dedicated GPUs slowed down training time, especially for EfficientNet-B3, which took 1 hour, 9 minutes, and 46 seconds, limiting our ability to explore hyperparameter combinations or try more complex architectures.

3. Trade-Offs

- a) Final model selection

EfficientNet-B3 was selected as the main model despite its longer training time, as it provided the most stable performance and highest accuracy. The trade-off was that prediction quality was more important than computational efficiency because the model would be used in real-world applications.

- b) Handling dataset imbalance

Actually, the dataset is not too imbalanced, so class weighting is used to handle this because it is faster and does not add computational load like data augmentation. However, class weighting has a trade-off, namely, suboptimal performance in classes with similar morphology, so that test accuracy does not increase significantly.

VI. CONCLUSION & FUTURE WORK

This study compares 3 fine-tuned CNN architectures, namely EfficientNet-B3, ResNet-50, and InceptionV3, for classifying dangerous farm insects. Based on the experimental results, the best model is EfficientNet-B3 with an accuracy of 0.79. However, the prediction performance is still lacking in four classes, namely Armyworms (F1-score 0.42), Fall Armyworms (F1-score 0.30), Corn Borers (F1-score 0.63), and Corn Earworms (F1-score 0.69). These four classes have morphological similarities and are in closely related genera, so the model often misclassifies them in real cases. Nevertheless, these results are still better than ResNet-50 and InceptionV3, because EfficientNet-B3 with its compound scaling is able to capture visual details more efficiently, so that the F1-score remains stable in most other classes. For future work, experiments with data augmentation should be considered to increase visual variation so that the model can learn more robust representations and reduce errors due to morphological similarities between classes.

VII. REFERENCES

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VIII. APPENDIX

1. Team Contribution Statement

Team Member	David Goanli	Kelson	Rafael Komala
EDA		✓	
Preprocessing			✓
Modelling	✓	✓	✓
Evaluation & Result	✓	✓	✓
Apps Deployment	✓		
Final Report		✓	✓
Demo Video	✓		
Presentation Slides		✓	✓

2. Supporting Documents

No.	Document	Link
1.	GitHub Repo	https://github.com/Luceatt/Final-Project-Deep-Learning
2.	Application	https://huggingface.co/spaces/DG226/DeepLearning
3.	Demo Video	OneDrive Demo Video
4.	Presentation Slides	Canva Presentation Slides

3. Screenshots and Code Snippets

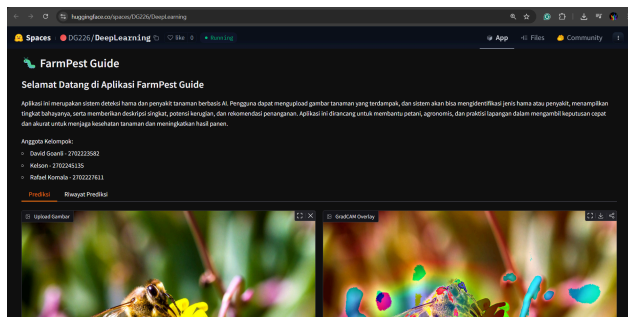


Figure 9. User Interface of the FarmPest Guide Application.

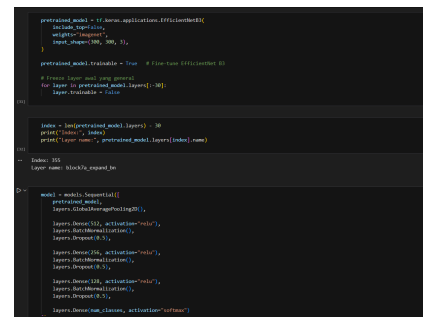


Figure 10. Code Snippet Example.